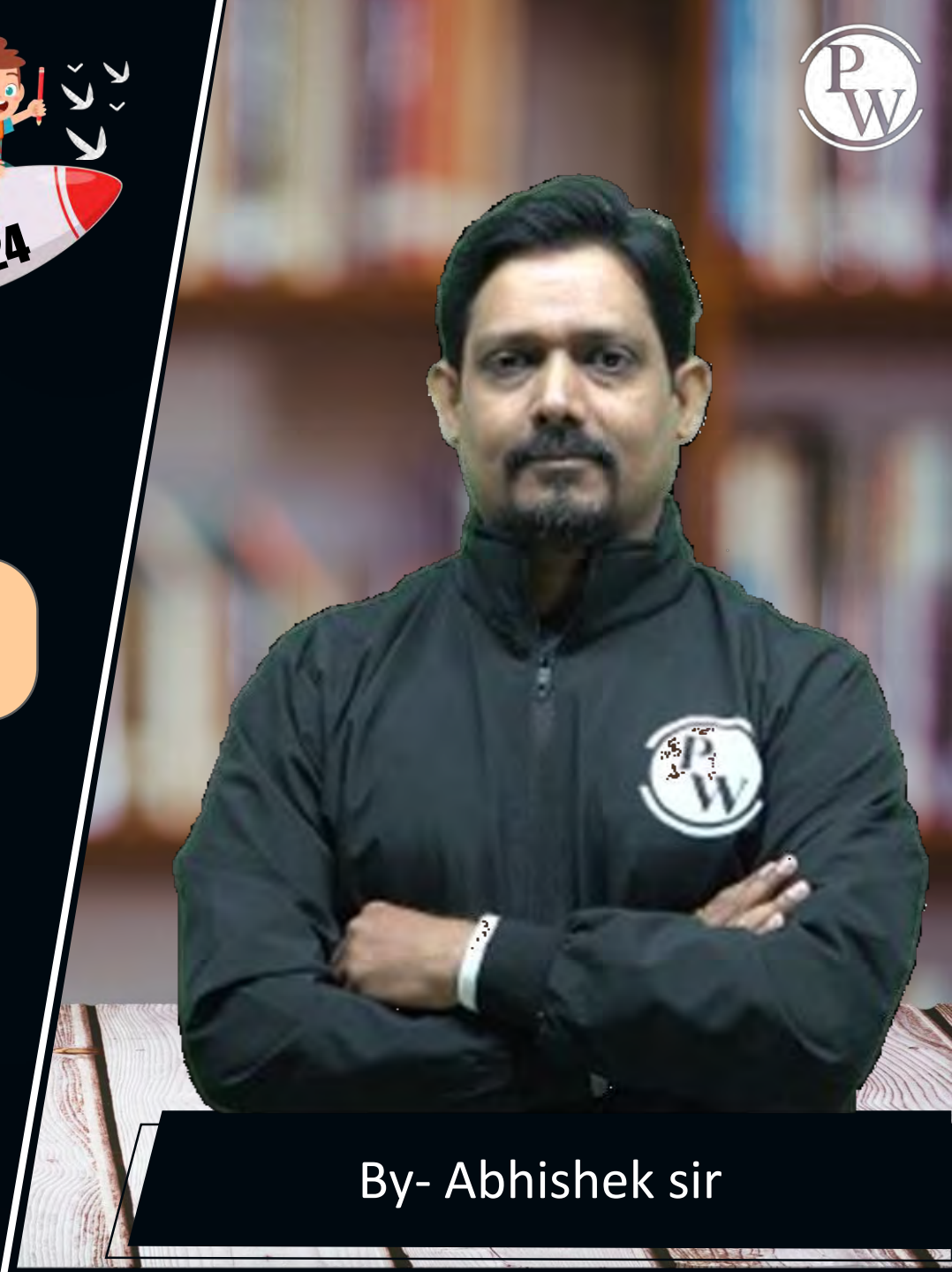
CS & IT ENGINEERING



C-Programming

Data type and Operators Discussion Notes

DPP - 1



By- Abhishek sir

[MCQ]

#Q.
```
include <stdio.h>
int main(void)
{
        int a;
        a = 2 * 6/5 + 3.0/2 + 1;
        printf("%d", a);
        return 0;
}
```

The value of a is 4

a = 12/5 + 3.0/2 + 1

= (2 + 1.5 + 1 = 4.5

= 3.5 + 1 = 4.5

A 4.9

B 4.0

C 4.5

D 4

[MCQ]

#Q.
```
#include <stdio.h>
int main(void)
{
        int a;
        a = 16.0 /4 * 5 % 3;
        printf("%d", a);
        return 0;
}
```
The value of a printed is ___

Division multiplication modules
Left to Right
16.0 /4 * 5 % 3 =
4.0 * 5 % 3
= 20.0 % 3
floating point
modulus
operator Not
allowed

A Compiler error ✓

B 8.0

C 2

D 8

[NAT]

#Q. Consider the following program.

```
#include<stdio.h>
void main()
 {
   int a;
   a = 32 > 24 > 13 > 10 > 8 > –1 > 0;
   printf("%d",a);
}
```

The output is ____1____.

Left to right

(((((32>24)>13)>10)>8)>-1)>0)

1 > 13

precedence

0 > 10 > 8 > -1 > 0

Associativity Rule

0 > 8 > -1 > 0

0 > -1 > 0

1 > 0 = 1

[NAT]

#Q.
```
#include<stdio.h>
void main()
{
    int a;
    a = 25 > 15 > 0! = 12 < 45 > 42! = 65;
    printf("%d", a);
}
```
The output is _____.

== , != lower precedence

25 > 15 > 0 != 12 < 45 > 42 != 65

1 > 0 != 12 < 45 > 42 != 65

1 != 1 > 42 != 65

(1 != 0 != 65)

1 != 65

1

[MCQ]

#Q. Consider the following program:

```
#include<stdio.h>
void main() {
   int a = 0, b = 1;
   a = (a = 5) && (b = 0);
   printf("%d", a);
   printf("%d", b);
}
```

The output is:

a = (a=5) && (b=0)

a = 0

A 50

B 00

C 10

D Compiler error

[MCQ]

#Q. Consider the following statements:

P: The precedence of the modulus operator is higher than multiplication or division operator.

Same - Incorrect

Q: The result of the modulus operator contains the sign of the second operand.

Which of the following statements is/are INCORRECT?

-5%2 → Negative

A Only P

B Only Q

C Both P and Q ✓

D Neither P nor Q

[MCQ]

#Q. Consider the following program:

```
#include<stdio.h>
void main()
{
   int a = 2022;
   printf("%d%d%d", a!=2024, a=2023, a==2021);
}
```

1 2023 0

The output is-

A 020220

B 020231

C 002021

D 120230

[NAT]

#Q. Consider the following program:

```
#include<stdio.h>
void main()
{
   int x=-2023;
   printf("%d", ~(x=x+5)) ;
}
```

The output is _____.

2017

6th operator

X = -2023

X = -2023 + 5 = -2018

~(-2018) ~2018 + 1

~ = -(-2018 + 1)

= -(-2017) = 2017

THANK - YOU